College of Engineering
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Software configuration management(SWEG5103)

SRS document for student online registration website for knowledge tribe company

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for the **Student Registration System (SRSys)** developed for **Knowledge Tribe Company**. It provides a complete and unambiguous description of system requirements in accordance with the **IEEE SRS standard**. This document serves as a reference for system design, development, testing, evaluation, and academic demonstration.

1.2 Document Conventions

- The term **"shall"** indicates a mandatory requirement.
- The term **"should"** indicates a recommended requirement.
- The term **"may"** indicates an optional requirement.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Project supervisors and evaluators (academic review)
- Developers and testers (implementation and verification)
- System administrators (deployment and operation)
- Stakeholders (requirements confirmation)

1.4 Product Scope

The **Student Registration System (SRSys)** is a web-based application designed **exclusively for student registration and record management**. It enables:

- Students to register online and maintain their personal data and documents
- Administrators to manage student records, documents, reports, and system settings securely

Out of scope: Teaching and learning management features, grading, course delivery, and payment processing.

1.5 Definitions, Acronyms, and Abbreviations

- **SRSys**: Student Registration System
- **CRUD**: Create, Read, Update, Delete
- **RBAC**: Role-Based Access Control
- **CSV**: Comma-Separated Values
- **SRS**: Software Requirements Specification

1.6 References

- IEEE 29148: Systems and Software Engineering — Life Cycle Processes — Requirements Engineering

1.7 Overview

This SRS includes system context, user classes, functional and non-functional requirements, external interfaces, database design overview, and future enhancements.

2. Overall Description

2.1 Product Perspective

SRSys is a standalone client-server web application based on a three-tier architecture:

- **Presentation Layer:** Web user interface
- **Application Layer:** Business logic and services
- **Data Layer:** Database system (MySQL or PostgreSQL)

2.2 Product Functions (High-Level)

The system provides:

- Online student registration
- Secure login and logout
- Student record management
- File upload and download
- Search, filtering, and sorting
- Email notifications
- Report generation and export
- Data exchange via RESTful APIs

2.3 User Classes and Characteristics

- **Administrator**
    - Manages student records, uploaded documents, reports, and system settings
    - Requires high privileges and secure access
- **Student**
    - Registers and updates personal information
    - Uploads required documents
    - Has restricted access to their own data only

2.4 Operating Environment

- **Client:** Modern web browsers (Chrome, Firefox, Edge)
- **Server OS:** Windows or Linux
- **Database:** MySQL or PostgreSQL

- **Email Service:** SMTP server/service

2.5 Design and Implementation Constraints

- Passwords shall be stored using secure hashing techniques.
- All user inputs shall be validated and sanitized.
- The system shall follow secure web development practices.

2.6 User Documentation

- Admin user guide (record management, reports, imports/exports)
- Student user guide (registration, profile updates, file uploads)

2.7 Assumptions and Dependencies

- Users have internet access.
- An SMTP email service is available for notifications.

3. System Features and Functional Requirements

3.1 Authentication and Authorization

Description: The system controls access using secure authentication and role-based permissions.

Functional Requirements:

- FR-1: The system shall allow users to register with valid credentials.
- FR-2: The system shall allow registered users to log in securely.
- FR-3: The system shall hash passwords before storing them in the database.
- FR-4: The system shall implement **role-based access control (RBAC)** for students and administrators.
- FR-5: The system shall manage user sessions using cookies.
- FR-6: The system shall allow users to log out and terminate active sessions.

3.2 Input Validation and Security

Description: The system validates and sanitizes all data received from users.

Functional Requirements:

- FR-7: The system shall validate all user inputs on the server side.
- FR-8: The system shall sanitize inputs to prevent SQL injection and XSS attacks.
- FR-9: The system shall display meaningful error messages for invalid input.

3.3 Student Record Management (CRUD)

Description: Administrators manage student records; students manage their own registration profile.

Functional Requirements:

- FR-10: The system shall allow administrators to create student records.
- FR-11: The system shall allow administrators to view student records.
- FR-12: The system shall allow administrators to update student records.
- FR-13: The system shall allow administrators to delete student records.
- FR-14: The system shall allow students to create and update their own registration data.
- FR-15: The system shall restrict students to accessing only their own records.

3.4 Search, Filter, and Sort

Description: Administrators can efficiently locate student records.

Functional Requirements:

- FR-16: The system shall allow administrators to search student records using multiple criteria.
- FR-17: The system shall allow administrators to filter student records.
- FR-18: The system shall allow administrators to sort student records by selected fields.

3.5 File and Document Handling

Description: Students upload documents; admins download and review.

Functional Requirements:

- FR-19: The system shall allow students to upload required documents.
- FR-20: The system shall allow administrators to download uploaded documents.
- FR-21: The system shall enforce file type restrictions.
- FR-22: The system shall enforce file size restrictions.

3.6 Email Service

Description: The system sends important notifications by email.

Functional Requirements:

- FR-23: The system shall send a confirmation email upon successful registration.
- FR-24: The system shall send notification emails for important updates.

3.7 Data Import and Export

Description: Administrators can import and export student data.

Functional Requirements:

- FR-25: The system shall allow administrators to import student data from CSV or Excel.
- FR-26: The system shall allow exporting student records and reports to CSV.
- FR-27: The system shall allow exporting student records and reports to Excel.
- FR-28: The system shall allow exporting reports to PDF.

3.8 Report Generation

Description: The system generates downloadable reports.

Functional Requirements:

- FR-29: The system shall generate student registration reports.
- FR-30: The system shall allow reports to be downloaded.

3.9 Date and Time Management

Description: The system stores consistent timestamps.

Functional Requirements:

- FR-31: The system shall store dates and times using a consistent standard format.
- FR-32: The system shall generate all timestamps automatically.

3.10 API Support

Description: The system exposes controlled access for integrations.

Functional Requirements:

- FR-33: The system shall provide RESTful APIs for authorized data access.
- FR-34: The system shall restrict API access to authorized users and roles.

4. External Interface Requirements

4.1 User Interfaces

- The system shall provide a responsive and user-friendly interface.
- Forms shall be clear, accessible, and validated.

4.2 Software Interfaces

- Database: MySQL or PostgreSQL
- Email: SMTP server/service

4.3 Communication Interfaces

- HTTPS shall be used for secure client-server communication.

- REST API communication shall use HTTP methods (GET, POST, PUT/PATCH, DELETE) with authentication.

5. Non-Functional Requirements

5.1 Performance

- NFR-1: The system shall support multiple concurrent users without performance degradation.

5.2 Security

- NFR-2: The system shall enforce secure authentication and authorization.
- NFR-3: The system shall protect session data from unauthorized access.

5.3 Reliability

- NFR-4: The system shall ensure consistent data storage and retrieval.

5.4 Maintainability

- NFR-5: The system shall be modular and well-documented.

5.5 Scalability

- NFR-6: The system shall support future functional extensions.

6. Database Design (High-Level)

6.1 Entities

- **Student** (StudentID, Name, DateOfBirth, Email, Phone, Address)
- **UserAccount** (UserID, Role, PasswordHash)
- **Document** (DocumentID, StudentID, FilePath, UploadDate)
- **Report** (ReportID, GeneratedDate)

6.2 Relationships

- One Student shall have one UserAccount.
- One Student may upload multiple Documents.

7. Future Enhancements

- Mobile-friendly application
- Advanced analytics dashboard
- Integration with external systems

8. Approval

This document represents the finalized Software Requirements Specification for the Student Registration System and is approved for academic submission and project demonstration.

10. Github link

https://github.com/Cep0110/knowledge-tribe